

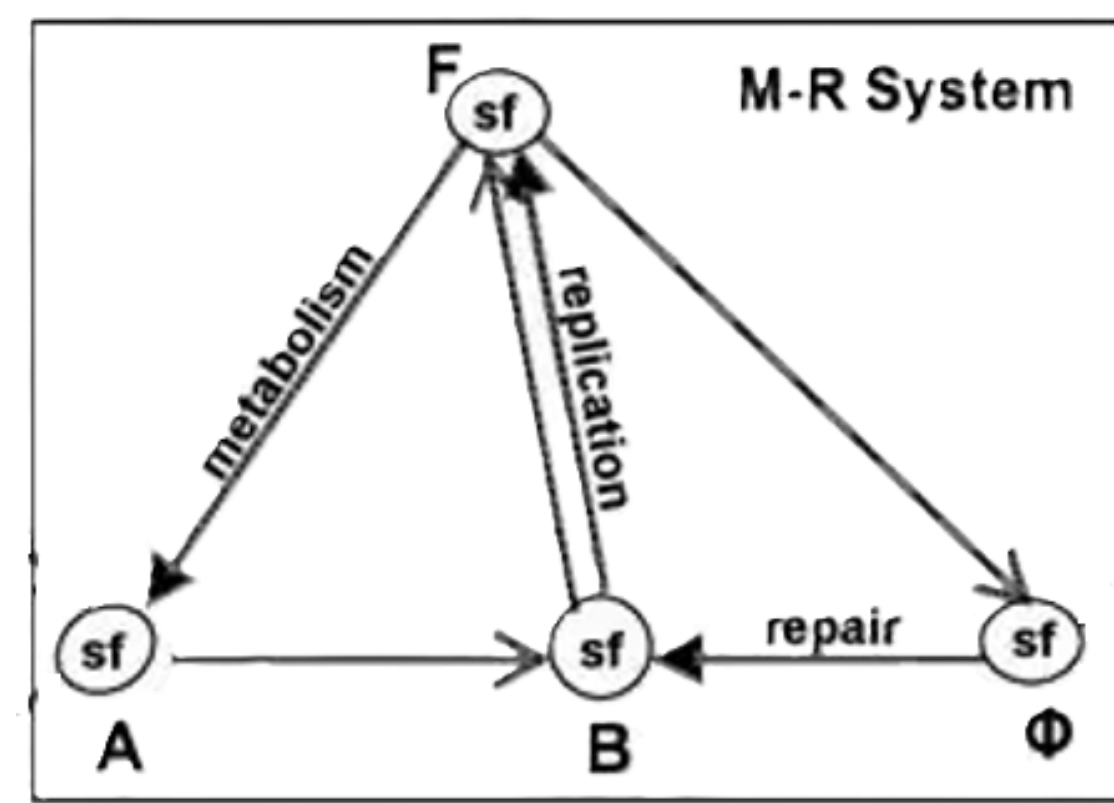
A New Grounding for (M,R) Systems

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1. (M,R) system in brief

Minimal model of life as **self-fabrication**, achieved by three sub-processes comprised of any number of “Actions” :

- Metabolism → synthesis
- Replacement → synthesis of synthesizers
- Replication → synthesis of replacement

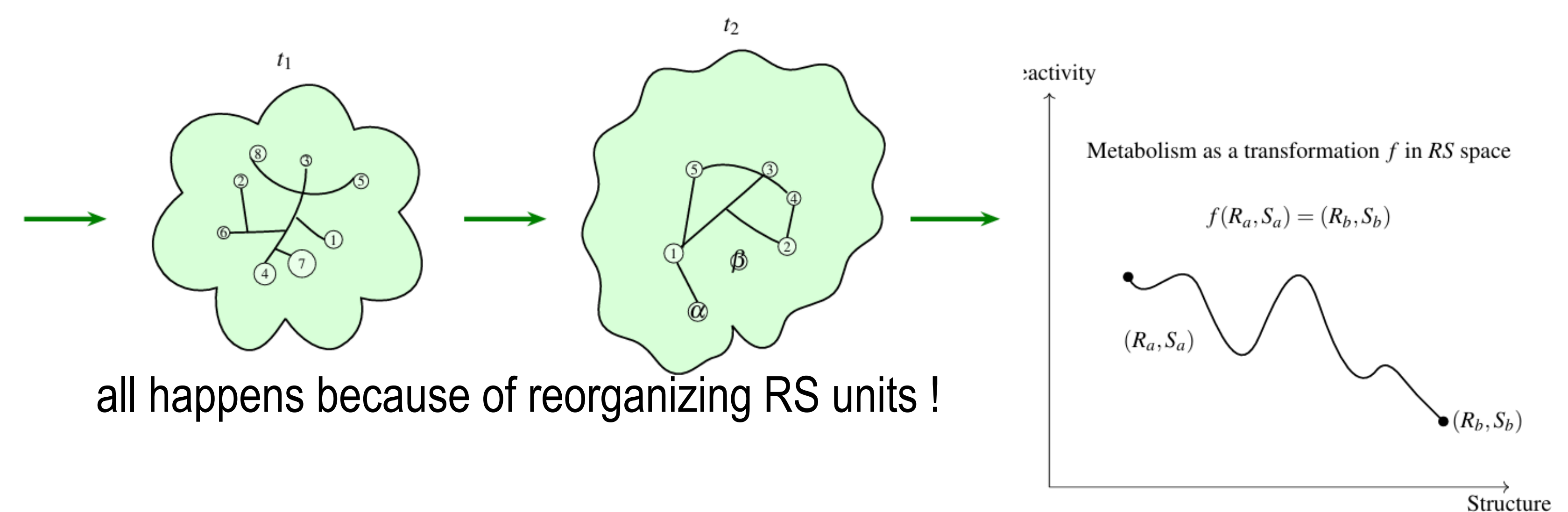


Grounding difficulties

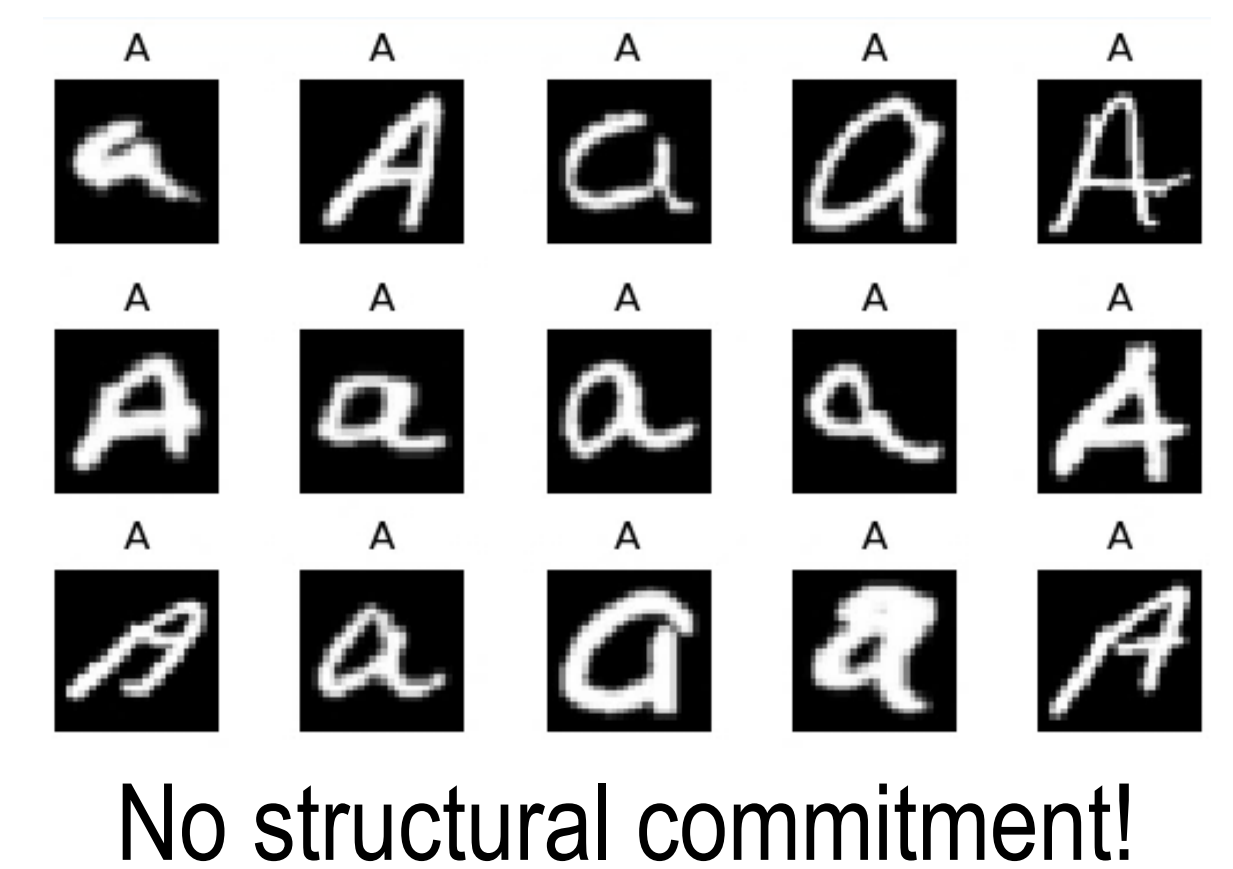
- Action = creation/synthesis : No extra semantics to map to Reality. No derivable “bio” properties e.g. homeostasis, evolution.
- Actions are Enzyme/Machine mediated, but not logically essential to the self-fabrication goal.

2. RS units : not all happens because of Enzymes

- Non-Enzymatic action is pervasive, at molecular level
 - Hofmeyr , FA systems and the Assembly action.
 - Can the theory express even more biological actions?
- **RS units** define material entities with two **intrinsic** properties : Reactivity + Structure
 - Reactivity → chemical interaction
 - Structure → shape and spatial occupancy
- Enzymes reduced to reactive and scaffolding RS units.
- All non-synthetic/non-covalent actions equally reducible
- We gained more plausible options to explain proto-life!

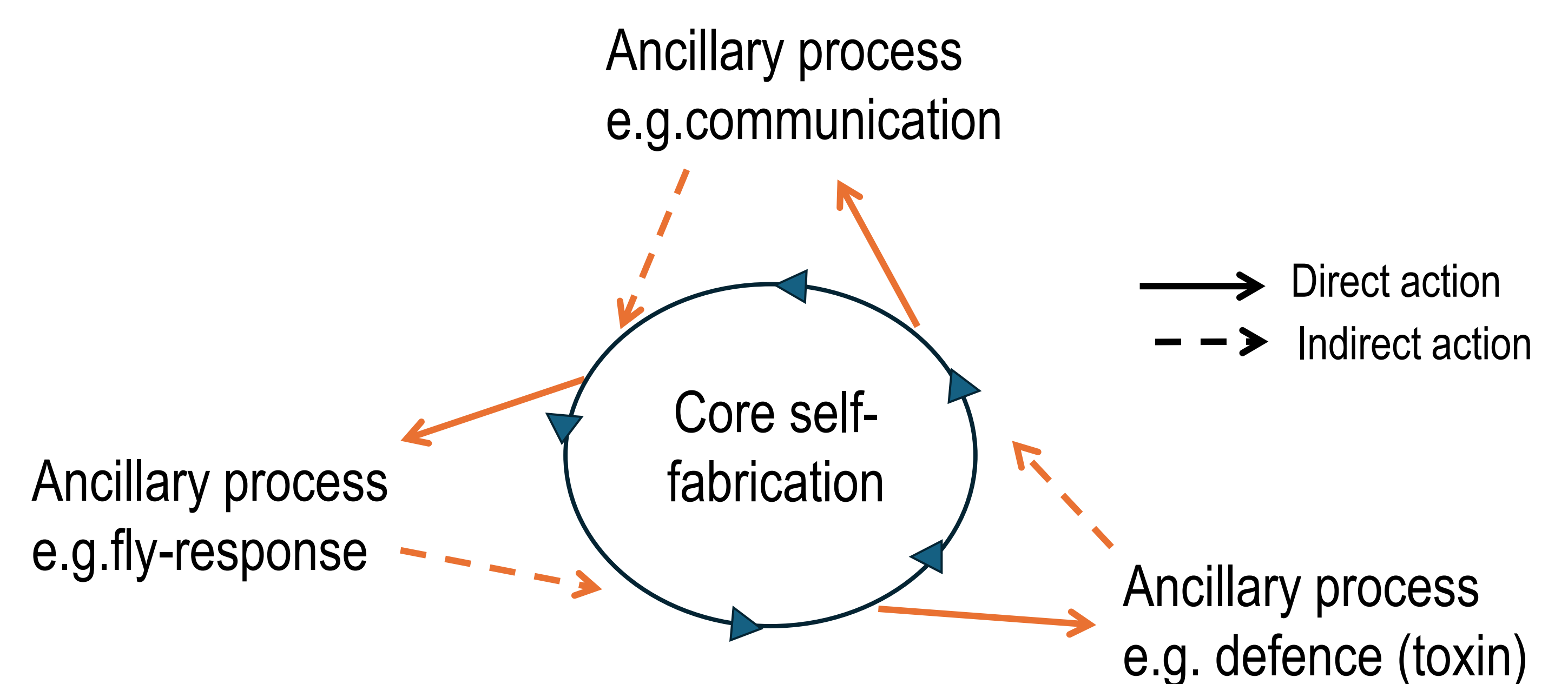


- RS units are no specific molecule, but a fundamental “placeholder” for the diverse role of matter in life.
- Formally, RS units improve conceptual separation between model design and contingent realisations.



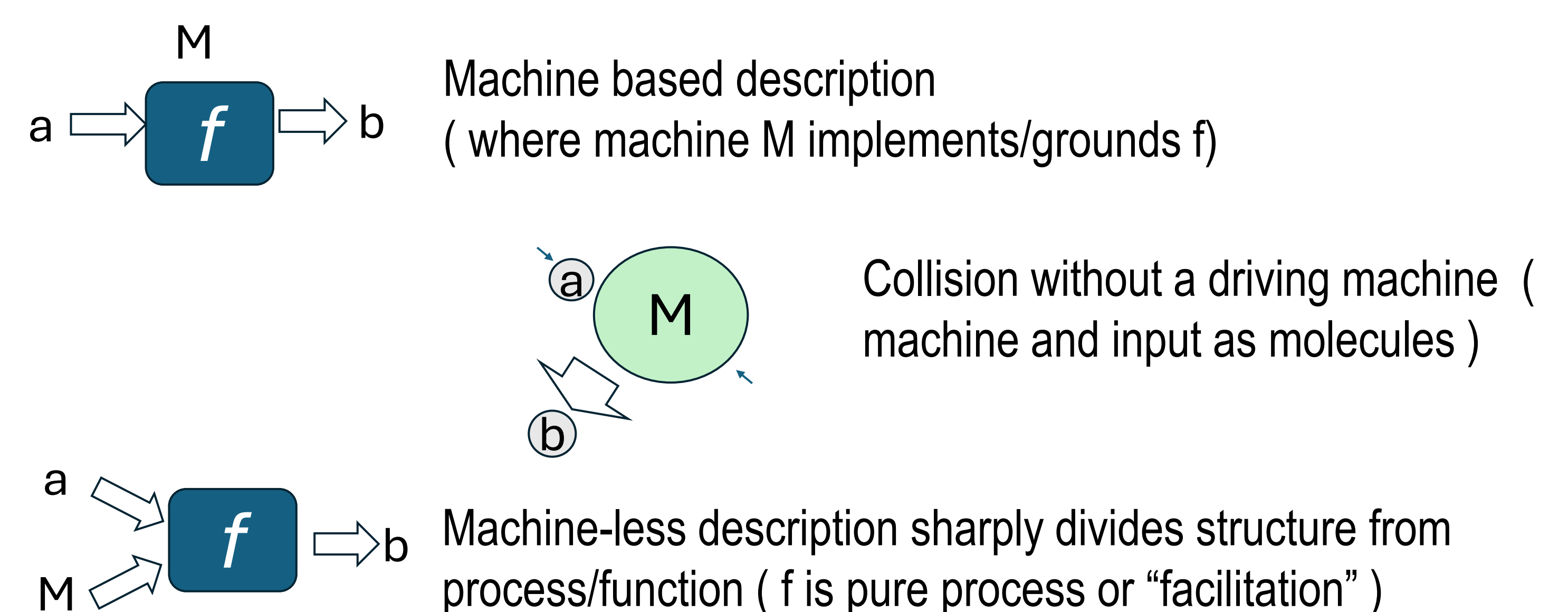
3. RS units : Broader functionality allowed

- Non-Enzymatic action is prevalent beyond molecular scale e.g. Cytoskeletal processes, physiology, biomechanics, etc. RS units remove the Enzyme constraint.
- Multi scale processes: all physics now “available for grounding” , and at any scale (cell , organism, ecosystem)
- **Persistence** becomes the broader goal : non-synthetic actions support self-fabrication, e.g. transport, signalling, defence, attack, and “behavioural” actions.



4. Did we break something?

- RS units **flatten the Machine/input hierarchy** (cartoon)
 - Enzyme action is simply a machine-less collision.
 - Yet, **without harming the basic self-fabrication goal!**
 - Perhaps is this a missing link towards $f(f) = f$?
- **“Function” (f) is more than Enzyme mechanism.** It is an arranged event of RS units, or “facilitation”.
- Roseannean causality closure : more than Enzyme creation, but perhaps hard to achieve in real systems.



5. ...a bold proposal at the end

- While self-fabrication is essential, the machine logic seems unnecessarily limiting.
- Persistence broadens the original self-fabrication model goal, and persistence and ancillary functions can complement